Roll No.: ....

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: B.Tech. CSE

Semester: III

Name of the Course: Mathematical Foundations for AI

Course Code: TCS 343

Time: 3 Hours

Maximum Marks: 100

Note:

i. **All questions are compulsory.**

ii. Answer any two sub questions among a, b & c in each main questions

iii. Total marks in each main question are twenty.

iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	Explain how word embeddings (like Word2Vec) form a vector space with meaningful geometric relationships.	CO1
(b)	Let $T: R^3 \to R^3$ be defined by $T(x,y,z) = (x+y, y+z, x+z)$. Find: The matrix representation; the kernel and image of T.	
(c)	Find a basis for the solution space of $x+y+z=0,\ x-y+2z=0$.	
Q2		
(a)	Consider a dataset where each data point is represented as a vector in R^3. The vectors are transformed using orthogonal projections and rotations during feature engineering. Describe how the Gram–Schmidt process can be applied to create an orthonormal feature basis for the dataset.	CO2
(b)	Why is the orthogonal complement important in Principal Component Analysis (PCA) and least squares regression?	
(c)	Apply Gram–Schmidt to $v_1 = (2,0,2)$, $v_2 = (3,3,0)$, $v_3 = (4,2,2)$ in R^3. Compute an orthonormal set.	
Q3		
(a)	Explain how singular value decomposition provides the best low-rank approximation of a matrix.	CO3
(b)	Find L such that $A = LL^T$, where $A = \begin{pmatrix} 25 & 15 & -5 \\ 15 & 18 & 0 \\ -5 & 0 & 11 \end{pmatrix}$.	CO4
(c)	Explain how eigenvectors can be used to find directions of maximum variance in data.	
Q4		CO5
(a)	Linearize the function $f(x,y) = x^2 + 3xy + y^2$ at the point (1,2).	
(b)	Explain why the maximum directional derivative occurs in the direction of the gradient vector.	
(c)	Describe how the directional derivative is used to verify correctness of gradients in backpropagation algorithms.	
Q5		
(a)	A spam filter classifies 80% of spam emails correctly and 90% of non-spam correctly. If 30% of emails are spam, find the probability that an email marked as spam is actually spam.	CO5
(b)	Discuss why the Gaussian (Normal) distribution is widely used for modeling noise in data.	CO6
(c)	Explain the role of prior, likelihood, and posterior in a Bayesian model.	